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ASML PhD Master Class Selection Committee

ASML Headquarters

Veldhoven, Netherlands

Subject: Application for the ASML PhD Master Class 2026

Dear ASML Selection Committee,

I would like to show my great interest for the ASML PhD Master Class 2026. Approaching the completion of my PhD in optical microscopy at the University of Innsbruck (Austria), I am looking forward to an environment where I can transition my technical expertise into industrial impact. I was familiar with ASML long before the current semiconductor boom, thanks to its collaboration with Zeiss and the fact that I studied in Jena, the city where Zeiss was grounded.

My doctoral research is a hands-on extreme precision measurements. I specialize in building custom optical setups, focusing on **Two-Photon Microscopy** and **Adaptive Optics**. With effort I have mastered the challenges of **femtosecond interferometry** by many hours in the dark lab. I understand that manufacturing nanometer-scale devices demands metrology with picometer-level precision and luckily a rigorous engineering challenge I face daily.

I find your vision inspiring by solving the humanity's toughest challenges. I would be flourishing with a professional environment that balances work stability with the freedom to imagine.

Throughout my career, I have cultivated a skill set aligned with modern lithography:

- **Extreme Precision Metrology:** Extensive experience constructing optical interferometers and overcoming stability challenges in femtosecond regimes and nanometer precision.
- **Wavefront Control & Simulation:** Deep understanding of phase manipulation by physical-mathematical models.
- **System Integration:** Proficient in automating experimental control and analysis pipelines using Python and LabVIEW-FPGA.
- **Communication:** Proven ability to disseminate complex research, evidenced by 5 peer-reviewed publications and presentations at 4 international conferences.

Outside the lab, I bring passion to my personal life, living in the Alps, I enjoy rides for **skiing and cycling**, and I also find joy in **dancing**. I believe this balance keeps my mind fresh and fosters the resilience needed for complex research.

I would appreciate your consideration of my extreme optimistic background, and I would be glad to be hear you back. Thank you for your time.

Sincerely,

Juan David Muñoz-Bolaños

R&D Optical Engineer- PhD

Medical University of Innsbruck